

Timothy C. Markfeld
Senior Software Engineer

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Senior Software Engineer with deep experience in .NET microservices, healthcare data integration, and AI-driven automation pipelines. Track record of building and owning production systems end-to-end across regulated and high-growth SaaS environments.

SKILLS

Languages C# (.NET) | Java | SQL | Python | JavaScript/TypeScript | Terraform
Technologies AWS | Entity Framework | GCP | Kafka | GraphQL | MassTransit | CI/CD | Datadog | Docker

PROFESSIONAL EXPERIENCE

Senior Software Engineer
FirmPilot

Apr 2025 – Present
Nashville, TN

- Built and owned fault-tolerant message-driven worker systems with retry logic and full observability supporting content, social media, and backlink automation workflows.
- Designed and implemented secure GraphQL APIs and React/Next.js TypeScript frontends with optimized database access patterns maintaining p99 query latency under 25ms across multi-tenant workloads.
- Engineered end-to-end AI-driven content pipelines integrating LLMs, fact-checking, media generation, and automated WordPress publishing.
- Led architecture cleanup and large-scale refactors, introducing CQRS patterns to improve long-term system maintainability.
- Merged 1,500+ PRs across backend, frontend, and infrastructure in under 14 months.
- Conducted daily code reviews and mentored engineers on architecture and best practices.
- Active on-call responder with 97.8% incident acknowledgment rate, 13s MTTA, and 22min MTTR across 45+ production incidents, with zero timeout escalations.
- Partnered with Product and Design to define technical requirements and drive architecture decisions.

Data Integration Engineer II
HCA Healthcare

Sep 2023 – Apr 2025
Nashville, TN

- Managed a system of 200+ HIPAA-regulated .NET microservices processing 1M+ daily events via Kafka and file-based ingestion across 10 SQL databases.
- Integrated with FHIR standards and external EHRs, ingesting and parsing HL7 messages for downstream clinical and operational workflows.
- Mentored and onboarded new engineers on legacy system architecture and HL7/FHIR standards.
- Architected and led migration of legacy microservices to Apache NiFi, reducing operational overhead and improving pipeline reliability.
- Implemented alerting and monitoring in Dynatrace with automated failover workflows.

EDUCATION

The University of Alabama

Sep 2019 – May 2023

- **M.S. Management Information Systems**, 4.0/4.0 GPA | **B.S. Management Information Systems**, 3.94/4.0 GPA

PROJECTS

Conductor: github.com/tcmarkfeld/Conductor

- CLI that generates least-privilege IAM policies and Terraform resources from MassTransit/SQS/SNS config for .NET teams deploying to AWS. Published on NuGet.

HL7Kit / HL7Kit.Fhir: github.com/tcmarkfeld/HL7Kit

- Strongly typed HL7 v2 parser with full segment coverage, dynamic delimiter handling, Z-segment support, and FHIR R4 conversion for .NET. Published on NuGet.

PriceTime: github.com/tcmarkfeld/PriceTime

- Price-time priority order matching engine for .NET with Kafka-driven order submission, trade execution, and real-time order book management.